Serverless Computing Meets LLM Inference

Current Research and Future Directions

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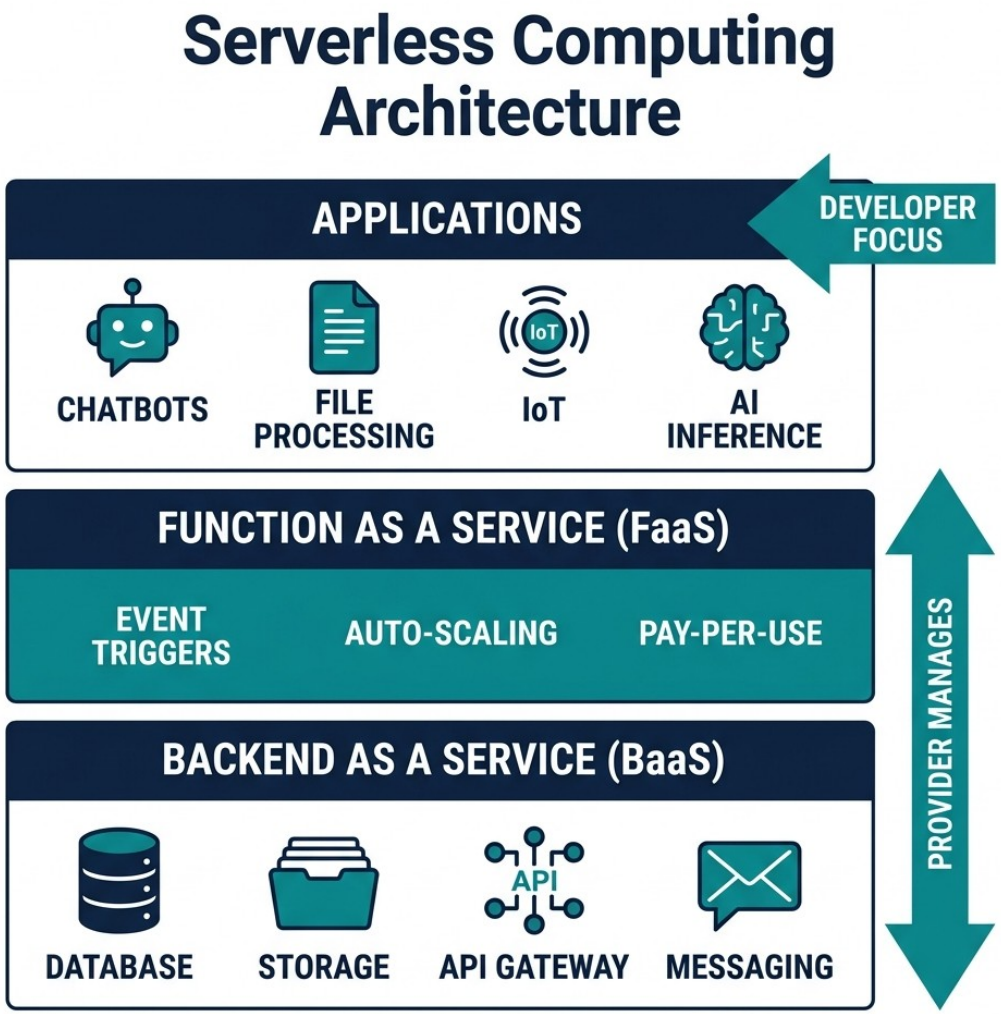
CS 6620 — Cloud Computing

Research Topic Presentation • Spring 2026

1 WHAT IS SERVERLESS COMPUTING?



Key Characteristics

- **Auto-scaling:** Scale up/down automatically on demand
- **Pay-per-use:** Charged only when your function runs
- **No server mgmt:** Provider handles all infrastructure
- **Event-driven:** Triggered by HTTP, queues, storage events
- **Stateless:** Ephemeral functions — no persistent state

Two Main Models

- FaaS — deploy code as event-driven functions
- BaaS — consume managed cloud services (DB, storage, auth)

2 ASSIGNED PAPER: SURVEY ON SERVERLESS COMPUTING

Hassan, Barakat & Sarhan (2021) — Journal of Cloud Computing • 275 papers reviewed (2016–2020)

Scope & Methodology

- **275 papers** systematically reviewed (2016–2020)
- **10 FaaS platforms** compared (Lambda, Azure, GCF, OpenWhisk, ...)
- **8 application domains:** chatbots, IoT, security, file processing, ...
- **Pricing models:** pay-per-invocation, per-duration, per-GB-second

📅 Timeline:
2014 — AWS Lambda launched • 2016 — Google & Microsoft adopt
2020 — Mature ecosystem with 10+ commercial platforms

Key Challenges Identified

- **Performance** (33 papers) — scheduling & call overhead
- **Cold Start Latency** (17 papers) — delay initializing idle functions
- **Security** (13 papers) — isolation & trust in shared infra
- **Vendor Lock-in** — strong dependency on provider ecosystem
- **Limited Duration** — short execution time limits
- **Stateless Design** — no persistent state across invocations

⚠️ **Critical Gap: No coverage of AI/ML/LLM workloads. This gap motivates the next two papers.**

3 TWO RESEARCH DIRECTIONS

The survey reveals a gap: no coverage of AI/LLM workloads. Recent research fills this gap in two directions.

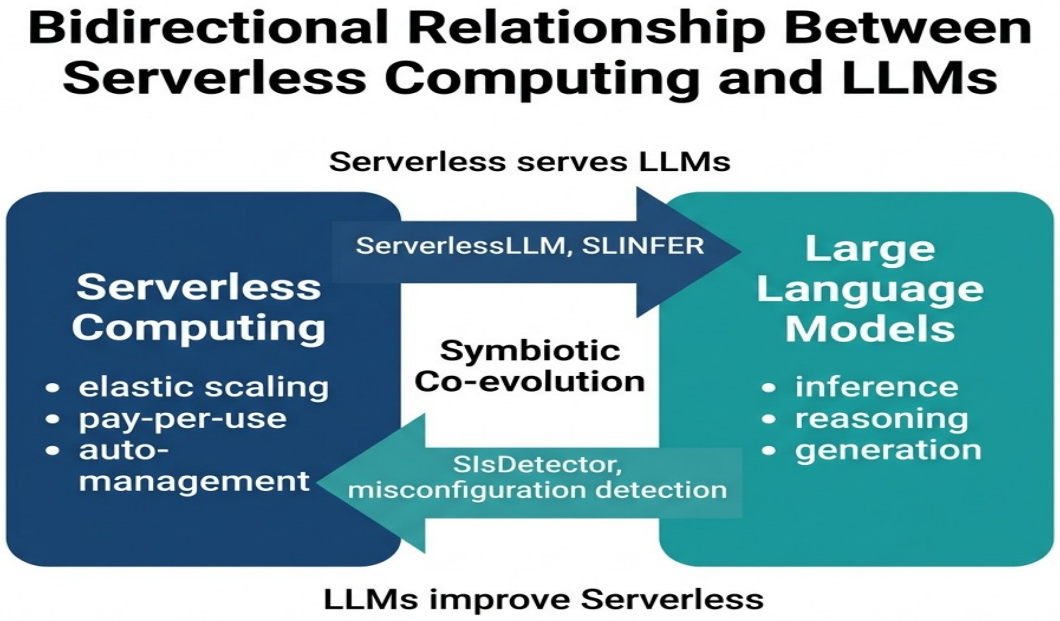
Direction 1

Serverless Serves LLMs

How do we deploy LLMs on serverless platforms?

- Cold start solutions
- Multi-tier storage
- Live migration

→ ServerlessLLM (OSDI '24)



Direction 2

LLMs Improve Serverless

How can LLMs make serverless more reliable?

- Misconfiguration detection
- Zero-shot prompting
- Chain-of-Thought reasoning

→ SlsDetector (TOSEM '26)

📋 **Presentation Structure: Survey Foundations → Direction 1: ServerlessLLM → Direction 2: SlsDetector → Open Questions**

DIRECTION 1: LLM INFERENCE ON SERVERLESS

Why deploy LLMs on serverless platforms?

Elastic Scaling

LLM workloads are extremely bursty — serverless scales automatically to handle spikes

Cost Efficiency

Pay only for active inference time — no cost for idle GPUs sitting between requests

Operational Simplicity

No GPU cluster management for developers — focus on the application logic only

But: LLMs break fundamental serverless assumptions

- **Cold start is catastrophic:** Loading a 70B-param model takes minutes — traditional serverless cold start is milliseconds
- **Stateless ≠ LLM inference:** KV-cache, conversation history, and multi-turn reasoning all require state across invocations
- **GPU resource mismatch:** Serverless platforms designed for CPU functions — GPU provisioning is coarse-grained and expensive

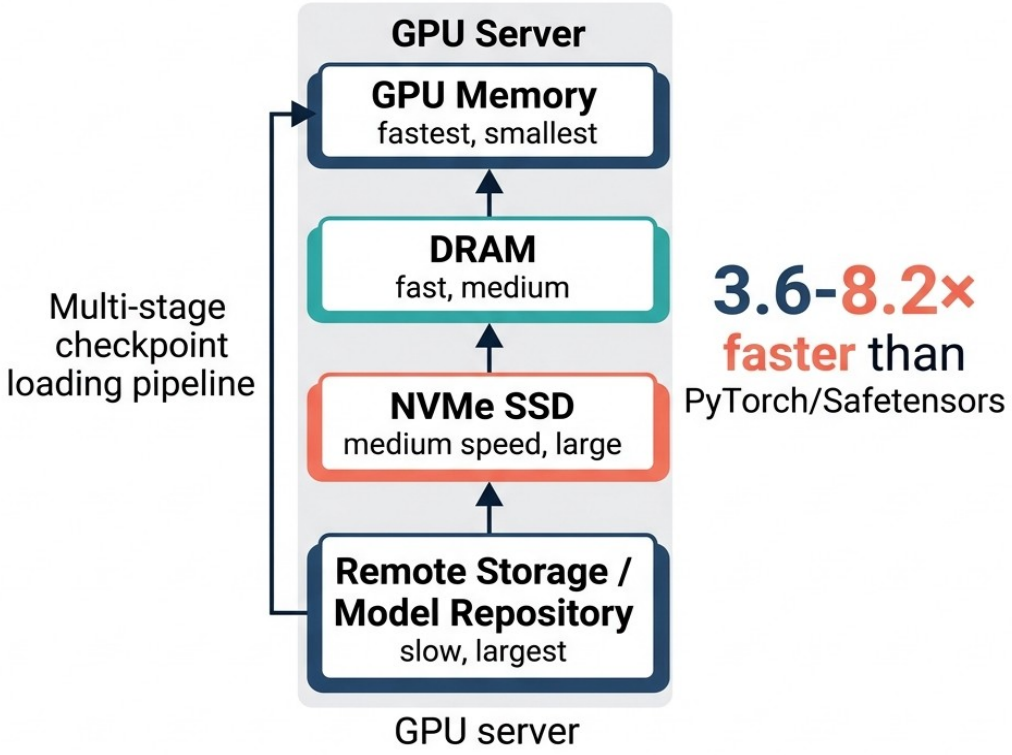
See also: Wang et al. (WoSC '24) — first survey of AI model inference on serverless, categorizing ML/DL/LLM workloads [4]

SERVERLESSLLM: SOLVING THE COLD START PROBLEM

"ServerlessLLM: Low-Latency Serverless Inference for Large Language Models"

Yao Fu, Leyang Xue, Yeqi Huang, A.-O. Brabete, D. Ustiugov, Y. Patel, Luo Mai • OSDI '24, USENIX, pp. 135–153, 2024

ServerlessLLM Multi-Tier Storage Architecture



Three Core Innovations

① Multi-Tier Checkpoint Loading

- Loading-optimized checkpoint format (sequential, chunk-based)
- Exploits full storage hierarchy: GPU → DRAM → SSD → remote
- Result: 3.6–8.2× faster than PyTorch/Safetensors

② LLM Live Migration (first ever)

- Migrates only tokens — NOT the massive KV-cache
- Re-computes KV-cache at destination, cutting network traffic

③ Locality-Aware Scheduling (Phantom)

- Cost models estimate load time per storage tier
- Picks optimal server to minimize startup latency

🚀 **10–200× latency improvement over KServe & Ray Serve (Azure Trace workloads)**

SERVERLESSLLM: EVALUATION & IMPACT

Fu et al. (2024) — OSDI '24 • Evaluated on GPU clusters with real-world workloads

Evaluation Setup

- **Hardware:** 8× A5000 GPUs, 1TB DDR4, NVMe SSDs (RAID 0)
- **Cluster:** 4 servers × 4 A40 GPUs, 10Gbps Ethernet
- **Models:** OPT (2.7B–66B), LLaMA-2 (7B–70B), Falcon (7B–40B)
- **Datasets:** GSM8K (math reasoning), ShareGPT (chat logs)
- **Workloads:** Azure Serverless Trace — bursty, real-world patterns

Key Results

3.6–8.2×
Faster checkpoint loading

10–200×
Latency improvement over baselines

4.4×
LoRA adaptor loading speedup

Why This Matters

- Proves serverless LLM inference is practical at production scale
- Multi-tier storage design is generalizable to any GPU cluster
- Live migration enables load balancing without dropping requests
- Directly addresses cold start — the #1 challenge from the survey

Related Recent Work

- SLINFER (HPCA '26): adds CPU+GPU heterogeneous sharing, achieving 86–154% capacity improvement [6]
- Samanta & Nguyen (WoSC '25): proposes 3-layer architecture vision for unified serverless LLM systems [5]

DIRECTION 2: LLMs IMPROVING SERVERLESS

"LLM-Based Misconfiguration Detection for AWS Serverless Computing"

Jinfeng Wen, Zhenpeng Chen, Zixi Zhu, F. Sarro, Yi Liu, Haodi Ping, Shangguang Wang • ACM TOSEM, 35(4), Article 110, 2026

The Misconfiguration Problem

- **800+ AWS resource types** with complex dependency chains
- **50,000+ IDs exposed** from a single S3 bucket misconfiguration
- **4.9M customers breached** from API misconfiguration (DoorDash)
- **Traditional methods fail:** data-driven approaches need large labeled datasets

SlsDetector: The LLM Approach

- **Zero-shot prompting** — no training data required at all
- **Chain-of-Thought** — step-by-step reasoning for complex configs
- **Multi-dim constraints** — aligned to serverless config semantics
- **Structured output** — deterministic, explainable detection

Results (ChatGPT-4o on 110 config files)

72.88%	88.18%	79.75%
Precision	**Recall**	**F1-Score**
+53.82 pp vs. SOTA	+17.40 pp vs. SOTA	+49.72 pp vs. SOTA

Generalizes Across LLMs

- ChatGPT-4o • Llama 3.1 (405B) Instruct
- Gemini 1.5 Pro • DeepSeek V3
- Consistent high effectiveness across all tested models

SYNTHESIS & OPEN RESEARCH QUESTION

The Story So Far

Survey (2021): Serverless is mature, but cold start, statelessness, and security are unsolved.

Gap: No coverage of AI/LLM workloads — the next frontier for serverless computing.

ServerlessLLM (2024): Solves cold start with multi-tier loading (10–200× improvement).

SlsDetector (2026): LLMs can improve serverless itself — bidirectional relationship.

Open Research Question

Can LLM agents autonomously manage serverless infrastructure?

Extending SlsDetector's approach from static detection to real-time autonomous infrastructure management.

Why This Question?

- SlsDetector proves LLMs understand serverless configs
- Auto-scaling, resource allocation, scheduling are all decision problems LLMs could reason about
- Would close the loop: LLMs serving on serverless, managed by LLMs — a self-optimizing system
- Combines cloud computing + AI — core CS 6620 themes

Key Takeaways

1. Serverless computing is evolving from general FaaS to AI-specific platforms
2. Cold start is the #1 barrier — ServerlessLLM achieves 10–200× improvement
3. The relationship is bidirectional: serverless serves LLMs, LLMs improve serverless
4. Open question: Can LLM agents autonomously manage serverless infrastructure?

Thank You — Questions?

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REFERENCES

[1] **Hassan, Barakat & Sarhan. "Survey on Serverless Computing."**
Journal of Cloud Computing, 10(39), 2021. [Assigned Paper]

[2] **Fu et al. "ServerlessLLM: Low-Latency Serverless Inference for LLMs."**
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[3] **Wen et al. "LLM-Based Misconfiguration Detection for AWS Serverless Computing."**
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[4] Wang, Jiang & Mi. "Advancing Serverless Computing for Scalable AI Model Inference."
WoSC '24, ACM, pp. 1-6, 2024.

[5] Samanta & Nguyen. "Illuminating the Hidden Challenges of Serverless LLM Systems."
WoSC '25, ACM, pp. 51-57, 2025.

[6] Xu et al. "Towards Resource-Efficient Serverless LLM Inference with SLINFER."
HPCA '26, IEEE, pp. 1-18, 2026.

Papers [1]–[3] are the core papers presented in depth. Papers [4]–[6] are supplementary references mentioned briefly.